Mohammad Hossein Amani

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PROFILE

Ph.D. candidate in Computer Science at EPFL (expected Jun 2027) working on reinforcement learning for reasoning in large language models. Research internships at Mistral AI and Apple on large-scale RL post-training. Recent papers at NeurIPS, ICLR and ACL.

EDUCATION

Ph.D. Computer Science Sep 2022 – Jun 2027 (expected)
École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland

B.Sc. Electrical Engineering and Physics (double major) Sep 2017 – Jun 2022
Sharif University of Technology, Tehran, Iran

EXPERIENCE

- **Research Intern, Mistral AI**, Paris — Oct 2026 – Present
  Developing on-policy self-distillation methods for large-scale RL post-training of LLMs
- **Research Intern, Apple Foundation Models**, Zurich — Apr 2026 – Sep 2026
  Trained 100B+-parameter foundation models with RL for multi-turn coding
- **RL Post-Training, Apertus Open LLM**, EPFL, Lausanne — Jun 2025 – Sep 2025
  Contributed to large-scale RL post-training of the Apertus open LLM

PUBLICATIONS

- On Reparameterizing the Score Function
  M. H. Amani, L. Bocchieri, M. Peyrard, C. Hongler, R. West — NeurIPS 2026
- RL for Reasoning by Adaptively Revealing Rationales
  M. H. Amani, A. Lotfi, N. M. Baldwin, S. Bengio, M. Farajtabar, E. Abbe, R. West — ICLR 2026
- Apertus: Democratizing Open and Compliant LLMs for Global Language Environments
  Apertus Team (contributed RL post-training) — ACL 2026
- Symbolic Autoencoding with Straight-Through Gradient Estimators
  M. H. Amani, N. M. Baldwin, A. Mansouri, M. Josifoski, M. Peyrard, R. West — ICML 2024 Workshop
- Sharp Asymptotics on the Compression of Two-Layer Neural Networks
  M. H. Amani, S. Bombari, M. Mondelli, R. Pukdee, S. Rini — IEEE ITW 2022
- Memorization and Optimization in Deep Neural Networks with Minimum Over-parameterization
  S. Bombari, **M. H. Amani**, M. Mondelli — NeurIPS 2022

AWARDS & SCHOLARSHIPS

- **EPFL Doctoral School Ph.D. Fellowship** 2022
- **Ranked 7th of 176 Electrical Engineering Students, Sharif** 2021
- **Ranked 370 of 140,000, National University Entrance Exam (Physics & Mathematics)** 2017
- **Silver Medal, Iranian National Physics Olympiad** 2016

SERVICE

- **Reviewer:** NeurIPS, ICML, ICLR

TEACHING

- **EPFL:** Machine Learning (2025), Applied Data Analysis (2023, 2024), Probability and Statistics (2023)
- **Sharif:** Probability and Statistics, High-dim. Probability, Signals and Systems, Programming (2018–2021)
- **Allame Helli High School:** Physics Olympiad coach (2017–2020)

SKILLS

- **LLM & RL training:** verl, TRL, DeepSpeed, FSDP, multi-node distributed training
- **Frameworks & languages:** PyTorch, Python, C++